Innate and learning in Language

Language competence

Innate in language: the inborn human capacity and constraints that enable language acquisition.

- Is language primarily the result of an innate cognitive endowment, or is it learned through Interaction with the environment?

Innate means *inborn*—present in the human mind before experience.

In linguistics, it refers to the **biological and cognitive capacities** that make language possible.

What is considered innate:

- The **capacity to acquire language**
- Sensitivity to **speech sounds**
- Ability to form **abstract grammatical categories**
- Understanding of basic **conceptual relations** (agent, action, cause, time)
- Inferential abilities used in **meaning interpretation**

 Innateness does **not** mean a child is born knowing a specific language, but born with a **language-ready mind**.

- **Innateness in Language**
- **2.1 Chomskyan Nativism**
- Noam Chomsky argued that humans are born with an innate **Language Faculty**, often formalized as **Universal Grammar (UG)**.
- **Key claims:**
- Language acquisition is **too rapid and uniform** to be explained by learning alone.
- Input available to children is **impoverished** (*Poverty of the Stimulus*).
- Therefore, core grammatical principles must be **biologically pre-specified**.
- **Innate components include:**
- Abstract syntactic categories (e.g., Noun, Verb)
- Structural principles (e.g., hierarchy over linear order)
- Constraints on possible grammars
- → Language learning, on this view, is **parameter setting**, not rule induction.

- **2.2 Biological and Cognitive Evidence**
- **Species specificity**: Only humans acquire full language.
- **Critical Period Hypothesis** (Lenneberg): Language acquisition is time-bound.
- **Neurobiological specialization**: Broca’s and Wernicke’s areas.

Learning in language: the process of acquiring a specific language through experience and interaction.

Learning in Language

Learning refers to what is **acquired through experience**, interaction, and exposure to a speech community.

What is learned:

- Vocabulary (words and expressions)
- Sounds and phonological patterns of a particular language
- Language-specific grammar
- Pragmatic norms (politeness, implicature, indirect meaning)
- Cultural and conversational conventions

 Learning happens through **use, interaction, and cognitive pattern-finding**.

Learning-Based Approaches

3.1 Behaviorism (Skinner)

Language is learned through:

- Imitation
- Reinforcement
- Habit formation

 Problem: Cannot explain creativity, novelty, or abstract structure.

3.2 Usage-Based and Emergentist Theories

Associated with **Tomasello, Goldberg, Bybee**.

Core ideas:

- Language emerges from **general cognitive abilities**.
- Grammar is learned from **usage patterns**.
- Constructions are learned pairings of **form and meaning**.

Example:

Children abstract grammatical patterns from frequent exemplars.

Here, **meaning, intention, and interaction** are primary.

3.3 Statistical and Computational Learning

- Learners track **probabilities and distributions** in input.
- Explains segmentation, category formation, and pattern generalization.

. Difference at a Glance

Innate	Learned
Biological endowment	Experience-based knowledge
Universal capacity	Language-specific systems
Cognitive constraints	Social and cultural norms
Makes language possible	Makes a language particular

- **How They Work Together**
- Language emerges from the **interaction** of both:
- **Innate mechanisms constrain and guide learning; learning fills in the linguistic content.**
- For example:
- A child is innately able to detect grammatical patterns,
- but must **learn Odia, English, or any other language** from input.

Innate

- **hardwired into the human brain** — part of our biological endowment.

Innate properties in language

- 1.**Universal Grammar (UG):**
- A set of structural principles shared by all human languages.
- Includes notions like recursion, hierarchical structure, and categories like noun and verb.
- 2.**Language Faculty (FLN** - Faculty of Language in the Narrow sense):
- Cognitive structures specialized for language processing.
- Includes the ability to acquire any natural language given minimal input (the "poverty of the stimulus" argument).
- 3.**Syntax-building capacity**:
- Abstract ability to generate grammatical sentences based on hierarchical rules, not just linear strings.
- 4.**Phonological capacity**:
- The ability to distinguish between speech sounds and organize them into a phonological system.
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Cultural and Learned Entities in Language

- These are **acquired through interaction, exposure, and social learning**, and vary widely across cultures and communities.

- 1. **Specific Languages** (e.g., Odia, Japanese, English):
  The vocabulary, grammar rules, pronunciation, and idioms are learned from the speech community.
- 2. **Lexicon** (words and meanings):
  While the capacity for storing words is innate, individual words and their meanings are culturally transmitted and learned.
- 3. **Writing systems**:
  Entirely cultural inventions — alphabets, scripts, spelling conventions, etc., are taught and not biologically innate.
- 4. **Pragmatic conventions:**
  Rules of politeness, turn-taking, indirectness, etc., vary by culture and are learned socially.
- 5. **Sociolinguistic variation:**
  Dialects, registers, styles of speech — influenced by social class, region, gender, etc.
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Summary in Table Form

Entity	Innate	Cultural / Learned
Universal Grammar	✔	☐
Syntax (as a capacity)	✔	☐
Phonological discrimination	✔	☐
Specific Language (e.g., Odia)	☐	✔
Vocabulary (Lexicon)	☐	✔
Writing System	☐	✔
Pragmatic Conventions	☐	✔
Sociolinguistic Variation	☐	✔

- The idea of **“copying” and the spread of cultural phenomena** explains **how learning operates on top of innate linguistic capacities**. In language, copying does not mean mechanical imitation alone; it is a **cognitively guided, socially mediated process** constrained by innateness.
- The innate language faculty enables every child to learn greetings, but the actual greeting is determined by culture.
- English: **Hello.**
- India: **Namaste.**
- Japan: **Konnichiwa.**
- Arab cultures: **As-salāmu ʿalaykum.**
- The capacity to learn greetings is innate; the specific form is culturally transmitted.

English word	Source language	Cultural significance
pizza	Italian	Italian cuisine
sushi	Japanese	Japanese food culture
curry	Tamil	South Asian cooking
taco	Spanish (Mexico)	Mexican cuisine
kimchi	Korean	Korean fermented food tradition
kebab	Turkish/Arabic	Middle Eastern cuisine

Example: We had **sushi** for

Innateness: What Is *Not* Copied

Innate aspects of language **do not spread by copying**.

These include:

- The **capacity for language**
- Sensitivity to structure and hierarchy
- Conceptual primitives (agency, causation, intention)
- Inferential abilities used in meaning interpretation

These are **biologically inherited**, not culturally transmitted.

 Therefore, **innateness sets the limits of what can be copied**.

Learning: Where Copying Operates

Copying is central to **learning and cultural transmission** in language.

2.1 Copying as Imitation

Children:

- Imitate **sounds, words, and constructions**
- Reproduce frequently heard patterns
- Copy intonation, rhythm, and pragmatic routines

Example:

A child copies the construction *“give me X”* before understanding abstract grammar.

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Copying as Pattern Replication

Copying is **selective and structured**, not blind.

Learners:

- Copy **recurring patterns**
- Generalize from exemplars
- Ignore noise and rare forms

This explains how:

- Idioms spread
- Slang emerges
- New constructions stabilize

Spread of Cultural Phenomena in Language

Language spreads culturally through **repeated copying across generations**.

Mechanisms:

1. **Social interaction** – conversation, storytelling, education
2. **Frequency effects** – frequent forms survive
3. **Prestige and authority** – elite or media forms spread faster
4. **Communicative efficiency** – easier forms are copied more

This aligns with:

- **Memetics** (Dawkins)
- **Cultural evolution models**
- **Usage-based linguistics**

Interaction of Innateness and Copying

Innateness **filters** what can be copied:

Thus, copying is **guided replication**, not free duplication.

Innate Constraint	Effect on Copying
Structural bias	Only human-possible grammars survive
Meaning bias	Forms with clear meaning spread
Inferential ability	Implicatures can be learned
Cognitive economy	Simplified patterns persist

Example from Language Change

- A new discourse marker (e.g., *like*, *you know*) appears
- Speakers copy it in similar contexts
- Children acquire it as part of grammar
- It becomes conventionalized

➡ **Cultural copying + innate constraints = language evolution**